

FLIGHT PROCEDURES MANUAL

PROJECT ZEPHYRUS

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Instructions

This manual is for use on Project Zephyrus flights, to be printed for launch in a minimum of 2 copies, one for the Telemetry Lead, and another for the Telemetry Deputy. It is to be read prior to launch by all members involved in the Ground Station, Away Station, and Avionics/Structures Lead roles. At each of the Away Station(s) and Ground Station, 2 copies must be printed and be available for the site lead and deputy to access prior to, during, and after flight. The personnel whose mastery of this document's contents is listed below:

- **Avionics Lead** (Sections 1-9)
- **Ground Station Lead** (Sections 1-2, 4-6, 8)
- **Ground Station Deputy** (Sections 1-2, 4-6)
- **Away Station Lead** (Sections 3-6, 9)
- **Away Station Deputy** (Sections 3-6, 9)
- **Simulation Lead** (Sections 1-10)
- **Structures Lead** (Sections 1, 4-6)

Contents

1	Avionics Setup Instructions	1	5.3	Flight Computer Reset Procedure	1
1.1	Avionics Bay	1	5.4	Lost Connection Procedure . .	2
1.2	Fin Control	1	6	Post-Flight Procedures	2
1.3	Airbrakes Module	1	7	Onboard Electronics Documentation .	2
1.4	Inter-Subsystem Connections	1	7.1	Onboard Hardware Documentation	2
2	Ground Station Setup Instructions .	1	7.2	Onboard Software Documentation	2
2.1	Telemetry Reception	1	8	Ground Station Documentation . . .	2
2.2	Live Video Reception	1	8.1	Ground Station Hardware . . .	2
3	Away Station Setup Instructions . .	1	8.2	Ground Station Software . . .	2
3.1	Antenna Pointer Assembly . .	1	9	Away Station Documentation	2
3.2	Antenna Pointer Setup	1	9.1	Away Station Hardware	2
4	Pre-Flight Procedures	1	9.2	Away Station Software	2
4.1	Go-No Go Criteria	1	10	FC Simulation Documentation	2
5	In-Flight Procedures	1	10.1	Usage Instructions	2
5.1	Expected Behavior	1	10.2	Usage Examples	2
5.2	Flight Detection Failure		10.3	FC Simulation Framework . . .	2
	Procedure	1	10.4	OR Simulation Framework . . .	2
			10.5	Python Scripting Framework .	2
			11	Notes from Xanthus Launch	2

1 Avionics Setup Instructions

1.1 Avionics Bay

1.2 Fin Control

1.3 Airbrakes Module

1.4 Inter-Subsystem Connections

2 Ground Station Setup Instructions

2.1 Telemetry Reception

2.2 Live Video Reception

3 Away Station Setup Instructions

3.1 Antenna Pointer Assembly

3.2 Antenna Pointer Setup

4 Pre-Flight Procedures

4.1 Go-No Go Criteria

5 In-Flight Procedures

5.1 Expected Behavior

5.2 Flight Detection Failure Procedure

TLDR advance state manually

5.3 Flight Computer Reset Procedure

TLDR assure visual apogee +1 second, arm and fire pyros manually; enter APOGEE state/DISREEF_AWAIT state manually

5.4 Lost Connection Procedure

TLDR cooked, antenna proc maybe

6 Post-Flight Procedures

7 Onboard Electronics Documentation

7.1 Onboard Hardware Documentation

7.2 Onboard Software Documentation

8 Ground Station Documentation

8.1 Ground Station Hardware

8.2 Ground Station Software

9 Away Station Documentation

9.1 Away Station Hardware

9.2 Away Station Software

10 FC Simulation Documentation

10.1 Usage Instructions

10.2 Usage Examples

10.3 FC Simulation Framework

10.4 OR Simulation Framework

10.5 Python Scripting Framework

11 Notes from Xanthus Launch

- * Possibly detect flight from remote rocket without human assistance
 - * FLIGHT -> FLIGHT, APOGEE will be detected asap anyway
 - * PREFLIGHT -> PREFLIGHT, ensure FLIGHT detection would work from in-flight
 - * MASTER PHYSICAL SWITCH: enable/disable set_state_from_SD_card on the rocket
 - * GROUND STATION SWITCH: if enabled, allows for <something>